

## Viscoelastic slender jets at $O(\varepsilon^2)$

Pre-requisites: This note continues *Viscoelastic slender jets* one order further in the slenderness parameter. It assumes that derivation: the parity argument, the leading-order axial equation, and the leading-order traction conditions. Primes are  $\partial_z$ , dots are  $\partial_t$ .

The leading-order slender-jet model closes with three fields,  $h$ ,  $v_0$  and the polymer normal-stress difference  $\Delta\Sigma$ . Going one order further tells us what the model leaves out and, more usefully, which of the neglected terms survive in specific geometries. The most important one is the radial pressure variation  $p_2$ : it carries the radial inertia of the thinning filament, which is the term that reappears in uniform-filament force balances (see *From the slender-jet hierarchy to a uniform-filament force balance*).

### Ordering

We use the axisymmetric, swirl-free jet in  $(r, z)$  with free surface  $r = h(z, t)$  and slenderness  $\varepsilon = \ell_r/L \ll 1$ . The bookkeeping is the standard one: each power of  $r$ ,  $h$ ,  $h'$  or  $h''$  carries one power of  $\varepsilon$ , while  $z$ -derivatives of the axial fields are  $O(1)$ .

There is no  $O(\varepsilon^1)$  correction: Axis regularity forces  $v_z$  and  $p$  to be even in  $r$  and  $v_r$  to be odd (see *why the expansions are in  $r^2$* ). The first neglected terms of the leading-order model are therefore  $O(\varepsilon^2)$ , not  $O(\varepsilon)$ .

### Field expansions

Velocity and pressure, now including the  $r^4$  terms:

$$v_z = v_0 + \varepsilon^2 r^2 v_2 + \varepsilon^4 r^4 v_4 + O(\varepsilon^6), \quad p = p_0 + \varepsilon^2 r^2 p_2 + \varepsilon^4 r^4 p_4 + O(\varepsilon^6),$$

$$v_r = -\frac{r}{2}v'_0 - \frac{\varepsilon^2 r^3}{4}v'_2 - \frac{\varepsilon^4 r^5}{6}v'_4 + O(\varepsilon^6),$$

where  $v_r$  follows exactly from *continuity* and axis regularity.

Polymer stress, consistent with *axis regularity of a tensor field*:

$$\sigma_{p,zz} = \Sigma_{zz} + \varepsilon^2 r^2 \Sigma_{zz,2}, \quad \sigma_{p,rr} = \Sigma_{rr} + \varepsilon^2 r^2 \Sigma_{rr,2}, \quad \sigma_{p,\theta\theta} = \Sigma_{rr} + \varepsilon^2 r^2 \Theta_2,$$

$$\sigma_{p,rz} = \varepsilon r S + \varepsilon^3 r^3 S_3.$$

The hoop and radial normal stresses share the same leading coefficient, so  $\sigma_{p,rr} - \sigma_{p,\theta\theta} = O(r^2)$ ; the difference  $\Sigma_{rr,2} - \Theta_2$  is a genuine second-order degree of freedom.

### Bulk equations

**Axial momentum** Collecting the  $r^0$  and  $r^2$  coefficients of the axial Navier-Stokes equation (the viscous radial Laplacian turns  $r^2$  into 4 and  $r^4$  into  $16r^2$ , see *the Laplacian in a slender jet*):

$$\rho (\partial_t v_0 + v_0 v'_0) = -p'_0 + \eta (4v_2 + v''_0) + \Sigma'_{zz} + 2S, \quad (\text{M0})$$

$$\rho (\partial_t v_2 + v_0 v'_2) = -p'_2 + \eta (16v_4 + v''_2) + \Sigma'_{zz,2} + 4S_3. \quad (\text{M2})$$

In (M2) the two cross terms  $v_r \partial_r v_z$  and  $v_2 v'_0$  cancel each other exactly, which is why the convective operator keeps the same form as in (M0).

**Radial momentum** At leading order the radial equation is satisfied identically:  $p_0$  has no  $r$ -dependence and a radial velocity linear in  $r$  has zero vector Laplacian. The first non-trivial radial balance is the coefficient of  $r^1$ , and it determines  $p_2$ :

$$p_2 = \frac{\rho}{4} (\partial_t v'_0 + v_0 v''_0) - \frac{\rho}{8} (v'_0)^2 - \eta \left( v'_2 + \frac{1}{4} v'''_0 \right) + \frac{3\Sigma_{rr,2} - \Theta_2 + S'}{2}. \quad (\text{R})$$

The inertial part deserves a closer look. Writing  $u_s = -\frac{h}{2}v'_0$  for the radial velocity of the interface, the term  $-\frac{\rho}{8}(v'_0)^2 r^2$  is the convective (Bernoulli-type) contribution  $\rho v_r \partial_r v_r$ , while  $\frac{\rho}{4} \partial_t v'_0 r^2$  is the local acceleration  $\rho \partial_t v_r$ . Keeping only the first of these is a common shortcut; the two are of the same order and, in a uniformly thinning filament, the  $\dot{h}^2$  parts cancel exactly.

**Kinematic condition**

At  $r = h$ , with  $v_z(h) = v_0 + \varepsilon^2 h^2 v_2$  and  $v_r(h) = -\frac{h}{2}v'_0 - \frac{\varepsilon^2 h^3}{4}v'_2$ ,

$$\dot{h} + (v_0 + \varepsilon^2 h^2 v_2) h' = -\frac{h}{2}v'_0 - \frac{\varepsilon^2 h^3}{4}v'_2 + O(\varepsilon^4),$$

which multiplies by  $2h$  into the conservative form

$$\partial_t (h^2) + \partial_z \left( h^2 v_0 + \frac{\varepsilon^2}{2} h^4 v_2 \right) = O(\varepsilon^4). \quad (\text{C})$$

The correction is the flux carried by the  $r^2$  part of the axial velocity profile.

**Traction conditions**

The geometry expands as

$$n = \frac{e_r - h' e_z}{\sqrt{1 + h'^2}}, \quad t = \frac{h' e_r + e_z}{\sqrt{1 + h'^2}}, \quad \kappa = \frac{1}{h} - \varepsilon^2 \left( h'' + \frac{h'^2}{2h} \right) + O(\varepsilon^4),$$

with the full mean curvature  $\kappa = \frac{1}{h\sqrt{1 + h'^2}} - \frac{h''}{(1 + h'^2)^{3/2}}$ . The sign of the second term matters: for a sphere  $h = \sqrt{a^2 - z^2}$  this expression gives  $2/a$ , while a plus sign gives zero.

**Normal traction** Solving  $n \cdot \sigma \cdot n = -\gamma \kappa$  for  $p_0$  through  $O(\varepsilon^2)$ :

$$p_0 = \frac{\gamma}{h} - \eta v'_0 + \Sigma_{rr} + \varepsilon^2 \mathcal{B} + O(\varepsilon^4), \quad (\text{N})$$

$$\mathcal{B} = -\gamma \left( h'' + \frac{h'^2}{2h} \right) - h^2 p_2 - \eta \left( \frac{3}{2} h^2 v'_2 + 4 h h' v_2 - h h' v''_0 - 3 h'^2 v'_0 \right) + \Delta \Sigma h'^2 - 2 h h' S + h^2 \Sigma_{rr,2}. \quad (\text{B})$$

Reading  $\mathcal{B}$  term by term: the curvature correction, the radial pressure non-uniformity evaluated at the surface, the viscous normal stress from the  $r^2$  velocity correction and from the slope of the interface, the projection of the polymer normal-stress difference onto the tilted normal, the polymer shear stress projected by the slope, and the  $r^2$  part of the polymer radial stress at the surface.

Tangential traction  $t \cdot \sigma \cdot n = 0$  gives, at leading order, the closure for  $v_2$  (as in the leading-order note):

$$\eta \left( -3h'v_0' - \frac{1}{2}hv_0'' + 2hv_2 \right) + hS - \Delta\Sigma h' = 0, \quad (\text{T0})$$

and at the next order the closure for  $v_4$ :

$$\eta \left( 4h^3v_4 - \frac{h^3}{4}v_2'' - \frac{7}{2}h^2h'v_2' - 2hh'^2v_2 + \frac{1}{2}hh'^2v_0'' \right) + h^3S_3 + h^2h'(\Sigma_{rr,2} - \Sigma_{zz,2}) - hh'^2S = 0. \quad (\text{T2})$$

(T2) is written without using (T0). If (T0) is used to eliminate  $hS - \Delta\Sigma h'$ , the Newtonian part takes the equivalent form  $-\frac{h^3}{4}v_2'' - \frac{7}{2}h^2h'v_2' - 4hh'^2v_2 + hh'^2v_0'' + 3h'^3v_0' + 4h^3v_4$  together with  $\Delta\Sigma h'^3$ , which is how it is often quoted.

#### Corrected axial equation

Insert (N) into (M0) and remove  $v_2$  with (T0). The polymer shear stress  $S$  cancels between the bulk term  $2S$  and the traction condition exactly as at leading order, and the result is

$$\boxed{\rho(\partial_t v_0 + v_0 v_0') = -\left(\frac{\gamma}{h}\right)' + 3\eta \frac{(h^2 v_0')'}{h^2} + \frac{(h^2 \Delta\Sigma)'}{h^2} - \varepsilon^2 \partial_z \mathcal{B}.} \quad (\text{E2})$$

The leading-order model is recovered by dropping the last term. Everything new at second order enters through the single scalar  $\mathcal{B}$ , which is why it is worth keeping  $\mathcal{B}$  as a named quantity.

#### Closed system

For the Newtonian hydrodynamic core, four fields close the problem at  $O(\varepsilon^2)$ :

- mass: (C) for  $h$ ,
- momentum: (E2) for  $v_0$ ,
- the  $v_2$  closure (T0),
- the  $p_2$  closure (R).

With polymer stresses, the same four equations hold with  $\Delta\Sigma$ ,  $S$ ,  $\Sigma_{rr,2}$ ,  $\Theta_2$ ,  $\Sigma_{zz,2}$  and  $S_3$  supplied by the constitutive equation expanded in  $r$  to the same order. (M2) and (T2) are only needed if  $v_4$  itself is wanted; they do not feed back into (E2) at this order.

#### What survives in a uniform filament

For a filament of uniform radius  $h(t)$  under homogeneous uniaxial extension,  $v_0 = -2(\dot{h}/h)z$ , one finds  $v_2 = v_4 = 0$ , the kinematic condition is satisfied exactly, and every term in  $\mathcal{B}$  vanishes except the pressure non-uniformity:

$$p_2 = -\frac{\rho}{2} \frac{\ddot{h}}{h}, \quad \mathcal{B} = \frac{\rho}{2} h \ddot{h}.$$

This single surviving term is the radial inertia of the thinning filament and is the origin of the inertial correction in uniform-filament force balances. The reduction is carried out in the next note.

#### • Related

- *Viscoelastic slender jets (leading order)*
- *From the slender-jet hierarchy to a uniform-filament force balance*
- *r-2-parity*
- *polymeric-stress-regularity-slender-jet*
- *dynamic-BC-slender-jet*